PTEN as the sampling reference: cost and equilibration metrics

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Abstract

The sampling reference changed from influenza HA0 (`zero_polymer`, $L = 566$) to human PTEN (`pten_1d5r`, $L = 324$, crystal 1D5R). This note collects what that costs and what it buys, with every number measured on astaroth (RTX 4500 Ada, 24 GB). Three things matter. A move is 1.37× cheaper, as the chain is shorter. Two chains now run on one card for 1.47× the throughput, where at $L = 566$ pairing was slightly *negative*: the shorter chain no longer saturates the GPU. And the reference-side chain reaches its H_d plateau in 10^4–2.5×10^4 moves, against $\sim4\times10^5$ for `zero_polymer` at a matched melted fraction, a factor of roughly 30. Taken together, a measurement that took a day now takes under an hour. The open problem has not gone away: below the transition the chains are trapped, and t_{eq} is reached at no temperature in the ladder.

1 Setup, and what is being measured

All numbers use the gradient-informed sampler with `grad_mode = one_pass` and one mutation per move ($N = 1$), the configuration the mutations-per-move scan selected (`DEVLOG` 2026–11–02: at fixed Δt the acceptance of an N-site move falls roughly like a_1^N while the cost of a move barely depends on N, so $N\cdot a$ peaks at $N = 1$ below the transition). The energy is the attention-map distance

$$U_{\mathrm{am}}(\sigma) \;=\; \sum_{i<j}\Big[\log A_{ij}(\sigma) - \log A_{ij}(\sigma_{\mathrm{ref}})\Big]^2, \tag{1}$$

with the reference map computed by ESM3 from the reference *sequence*, so $U_{\mathrm{am}}(\sigma_{\mathrm{ref}}) = 0$ by construction and no experimental structure enters. Two consequences are used below: U_{am} is a sum over residue *pairs*, so the energy scale of one mutation grows roughly like L^2; and H_d, the Hamming distance to the reference, runs from 0 to L.

Hardware and software are unchanged between the two references, so every comparison below is a comparison of chain length, not of machine.

2 Cost of one move

The one-pass move is worth 2.36× at $L = 324$ against 2.38× at $L = 566$: **the speed-up does not depend on chain length**, so the shorter reference costs the method nothing. What length changes is the absolute cost, 0.1118 against 0.153 s, a factor 1.37.

3 Cost of 10^5 moves

The same table read as wall-clock hours for 10^5 accepted-or-rejected moves of *aggregate* sampling, which is the unit a run is actually planned in.
Between the worst row and the best there is a factor 4.8, and 2.9 of it is free: the one-pass move and running two chains instead of one.

reference	grad_mode	s/move, one job	s/move, two jobs	effective interval
pten_1d5r ($L = 324$)	one_pass	0.1118	0.1516	**0.0758**
pten_1d5r ($L = 324$)	site_soft	0.2632	—	0.2632
zero_polymer ($L = 566$)	one_pass	0.153	0.330	0.165
zero_polymer ($L = 566$)	site_soft	0.365	0.79	0.395

Table 1: Seconds of wall clock per move. "Two jobs" is the per-chain figure with two independent chains sharing the card; "effective interval" is the wall time per move produced in total, i.e. half the two-job figure. PTEN numbers from `onepass_study_pten/analysis/bench.tsv` (300 moves at $T = 100$) and from the 2026–11–12 ladder; `zero_polymer` from `DEVLOG` 2026–10–22 and 2026–11–02 and the 2026–11–11 ladder.

reference	configuration	h per 10^5 moves	moves per hour
pten_1d5r	one_pass, two chains	**2.11**	**47 500**
pten_1d5r	one_pass, one chain	3.11	32 200
pten_1d5r	site_soft, one chain	7.31	13 700
zero_polymer	one_pass, one chain	4.25	23 500
zero_polymer	one_pass, two chains	4.58	21 800
zero_polymer	site_soft, one chain	10.14	9 900

Table 2: Derived from Table 1. Best configuration per reference in bold for PTEN; note that for `zero_polymer` the best is *one* chain.

4 Two chains on one card, which pays only now

At $L = 324$ a single job leaves the GPU 41 % idle. Pairing therefore buys throughput, which it did not at $L = 566$.

	one job	two jobs	change
s/move, each chain (PTEN)	0.1118	0.1516	
effective interval (PTEN)	0.1118	0.0758	1.47× throughput
GPU utilisation (PTEN)	59 %	92 %	
card memory in use (PTEN)	5514 MB	10 773 MB	
peak allocated per job (PTEN)	4327 MB	4327 MB	
effective interval (`zero_polymer`, `one_pass`)	0.153	0.165	8 % *worse*
peak allocated per job (`zero_polymer`)	6350 MB	—	

Table 3: Concurrency at the two chain lengths. The sign of the effect reverses with length, so `MAX_RUNNING=2` is right for PTEN and wrong for `zero_polymer`.

Two practical consequences. First, 10 773 MB of 24 GB means a third concurrent chain would fit, and whether it pays has not been measured. Second, **s/move is only comparable between runs at the same `MAX_RUNNING`**: the 0.33 s/move in the 2026–11–11 `zero_polymer` ladder and the 0.153 s/move single-job figure are the same sampler.

5 Equilibration

The ladder of 2026–11–12: nine temperatures, two chains each (one started at the reference, one at a random canonical sequence), 3×10^4 one-pass moves per chain, 5.4×10^5 moves in all, 11.5 h of card time. Per chain, over the second half of each run (`pten1d5r_study/analysis/`):

	started at the reference			started at random		
T	$\langle U \rangle$	$\langle H_d \rangle$	acceptance	$\langle U \rangle$	$\langle H_d \rangle$	acceptance
1	45	18	0.0097	46 691	303	0.0018
2	304	74	0.0552	43 838	305	0.0033
4	996	162	0.1234	44 694	305	0.0011
8	1879	215	0.2138	43 636	308	0.0019
16	2870	246	0.2950	45 690	307	0.0037
32	4220	267	0.3631	44 915	307	0.0105
64	5799	272	0.4516	42 048	310	0.0581
128	9000	285	0.5238	45 163	305	0.3717
316	16 324	289	0.5736	39 279	305	0.6137

Table 4: The chain-averaged table that `analyze_nsites.py` printed before 2026–11–12 hid all of this: its $\langle H_d \rangle = 268$ at $T = 1$ is the average of 18 and 303, a value neither chain ever visits. The per-chain block is now printed under every temperature.

5.1 How long the reference-side chain takes to plateau

H_d of the reference-started chain, averaged over 5000-move blocks:

T	H_d per 5000-move block	plateau	moves to plateau
4	66 / 121 / 143 / 158 / 168 / 160	≈ 160	$\sim$20 000
8	125 / 205 / 211 / 213 / 220 / 213	≈ 213	$\sim$12 500
16	177 / 242 / 240 / 243 / 249 / 246	≈ 246	$\sim$10 000
32	214 / 270 / 268 / 273 / 265 / 262	≈ 267	$\sim$10 000

Table 5: U plateaus in the same window. $T = 1$ and $T = 2$ are omitted deliberately: see the caveat below.

The comparison with `zero_polymer` has to pair temperatures by *melted fraction*, not by the number T, because the two references have different L and therefore different energy scales. PTEN at $T = 8$ plateaus at $H_d/L = 213/324 = 66\,\%$; `zero_polymer` at $T = 12$ plateaus at $380/566 = 67\,\%$. Those are the comparable runs, and `zero_polymer`'s took about 4×10^5 moves to get there: a fast rise to $\sim 2 \times 10^5$ at $H_d = 346$, a step to $\sim$380 near 4×10^5, then flat over the last $\sim 1.5 \times 10^5$ of a 5.8×10^5-move run (`DEVLOG` 2026–11–02).

reference	matched run	moves to plateau	card time
`pten_1d5r`	$T = 8$, 66 % melted	$\sim$12 500	**0.5 h**
`zero_polymer`	$T = 12$, 67 % melted	$\sim$400 000	17 h

Table 6: Card time at each reference's best configuration (PTEN two chains at 0.1516 s/move each, `zero_polymer` one chain at 0.153 s/move). The factor is about 30 in moves and about 30 in time; pairing other temperatures instead gives 16–40, so the honest claim is "an order of magnitude", not a precise number.

6 Where the transition is

$\langle H_d \rangle$ of the reference-side chain runs 18, 74, 162, 215, 246, 267, 272, 285, 289 over $T = 1 \ldots 316$ with no jump, and acceptance rises smoothly from 0.0097 to 0.57. Halfway to a random sequence ($H_d \approx 307$) falls at $T \approx 4$, so the crossover is at $T \approx$ 3–5, against $\sim$8–18 for `zero_polymer`. That

is what the pair-sum structure of U_{am} predicts: the energy per mutation scales roughly like L^2 and $(324/566)^2 = 0.33$, so PTEN's crossover should sit a few times lower, and it does.

7 Caveats

1. t_{eq} **is reached at no temperature** in this ladder, by the criterion in `analyze_nsites.py` ($\hat{R} < 1.1$ on both U and H_d from t_{eq} to the end). The plateaus above are one-sided: they say the reference-started chain has stopped moving, not that it samples the equilibrium distribution.

2. $T = 1$ **and** $T = 2$ **are flat because they are trapped**, not equilibrated: 207 and 1425 accepted moves in 3×10^4. Only $T \geq 4$ has thousands of accepted moves and a plateau worth the name.

3. **The random-started chain is useless below the crossover**, exactly as on `zero_polymer`: $U \approx 42\,000$–$47\,000$ at every $T \leq 64$ and H_d never leaving 303–310. At $T = 64$ it fell from 48 285 to 41 183 over 3×10^4 moves with ∼23 000 still to go. So $\hat{R}$ across opposite starts compares two basins instead of measuring one. `study_nsites.sh` now takes `STARTS`, and `STARTS="reference reference"` gives two reference-side chains differing only by proposal seed: a weaker diagnostic, but one about the basin being measured.

4. H_d **to the reference stops being a convergence coordinate above the crossover.** $\hat{R}_{H_d}$ sits at 3–9 for every $T \geq 16$ and looks converged while $\hat{R}_U$ is 10–150 on the same rows. A random canonical sequence is $19/20 = 95\,\%$ different from any reference, so once the reference-side chain melts past $H_d \approx 250$ both chains are near 300 and H_d cannot separate them. U can: at $T = 316$ the two are at 16 324 and 39 279. Read $\hat{R}_U$.

5. A third concurrent chain fits in memory and has not been timed.

6. The mutations-per-move choice $N = 1$ is carried over from `zero_polymer` and has not been re-measured on PTEN.

8 Reproducing the numbers

Per-chain tables, the $\hat{R}$ series and the per-temperature plots of U and H_d (one curve per chain, with t_{eq} marked):

```
cd softmax/monte_carlo
python analyze_nsites.py equil $PWD/pten1d5r_study --every 1000
```

It writes `pten1d5r_study/analysis/equil_T<T>.tsv` and `equil_T<T>.png` and needs only numpy and matplotlib, no GPU. The move-cost benchmark is `study_onepass.sh` with `REF=pten_1d5r`, analysed by `analyze_onepass.py`; the ladder is `study_nsites.sh` with `REF=pten_1d5r TS_OVERRIDE="1 2 4 8 16 32 64 128 316" NCHAINS=2 MOVES_EQ=30000 GRAD_MODE=one_pass MAX_RUNNING=2`.